Reconeyez Integration and Alarm Configuration Guide

Table of contents

- Introduction
-
- System Architecture
  - **Field Layer – Detection Network**
  - **Communication Layer – Bridge Connectivity**
  - **Cloud Layer – Reconeyez Cloud Platform**
  - **Integration Layer – Genesis NXG**
  - **Monitoring Layer – Command Center**
  - **End-to-End Flow Summary**
- Configuration Overview
  - Configuration in Reconeyez Cloud
    - Information Required by Reconeyez Support
    - Area Assignment in Reconeyez Cloud
  -
  - Configuration in Genesis
    - Device Discovery and Account Mapping in Genesis
-
- Integration Features Overview
- Summary

Introduction

When we refer to *Reconeyez Monitoring* in Genesis, we are describing a cohesive integration of wireless, battery-powered security devices designed for remote surveillance and intrusion detection.
Reconeyez systems combine intelligent motion sensors, image verification, sirens, and communication bridges to provide autonomous perimeter security even in areas with limited infrastructure or unstable power and network conditions.

Each device — whether it's a **detector**, **siren**, or **bridge** — plays a critical role in ensuring continuous situational awareness. Together, they form a distributed wireless mesh network that communicates securely with the Reconeyez Cloud and forwards verified alarms to the Genesis platform.

This integration enables **event-based video verification**, **low-power operation**, and **real-time alarm delivery**, making Reconeyez ideal for monitoring remote towers, temporary construction zones, and high-value assets in off-grid environments.

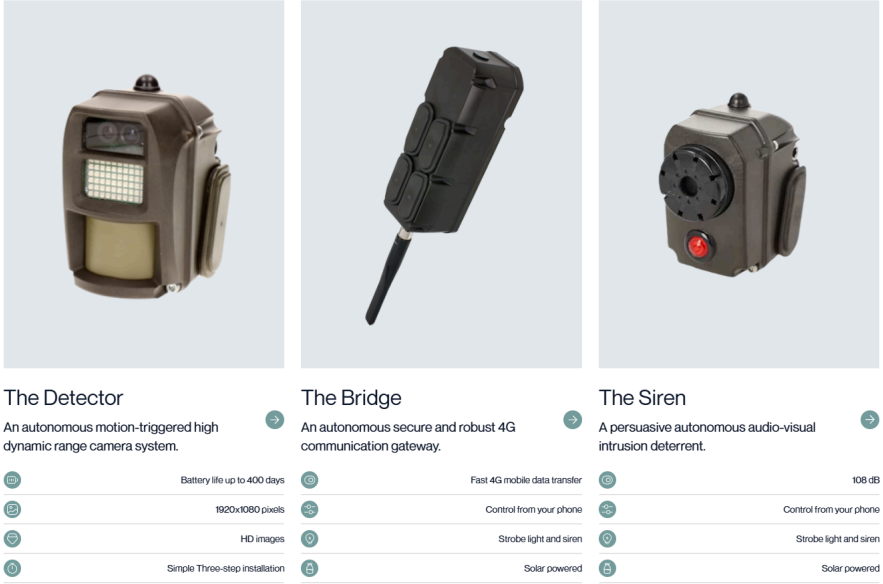
System Overview

A standard Reconeyez deployment comprises three core components:

- **Detectors (D)**
  Smart PIR-based motion sensing camera equipped with image capture capability. They trigger visual verification snapshots upon detecting movement within the configured detection zone.

- **Sirens (S)**
  Battery-powered siren units that provide local audible deterrence. They are wirelessly linked to detectors and bridges for coordinated alarm response.

- **Bridges (B)**
  Communication hubs that aggregate wireless events from nearby detectors and sirens and forward them securely over 4G or I/O interfaces to the Reconeyez Cloud.



Each device communicates via a long-range proprietary radio protocol, allowing wide-area coverage with minimal infrastructure.
The **Bridge** acts as the central gateway, connecting the local Reconeyez network to the Genesis platform through the Reconeyez Cloud API.

Within Genesis, all telemetry and image-verification events received from the Reconeyez Cloud are automatically classified, timestamped, and mapped to the corresponding monitored site.

This architecture ensures:

- Real-time visibility into motion events and verified images.
- Centralized alarm handling with event confirmation.
- Automatic status synchronization between the field devices and Genesis

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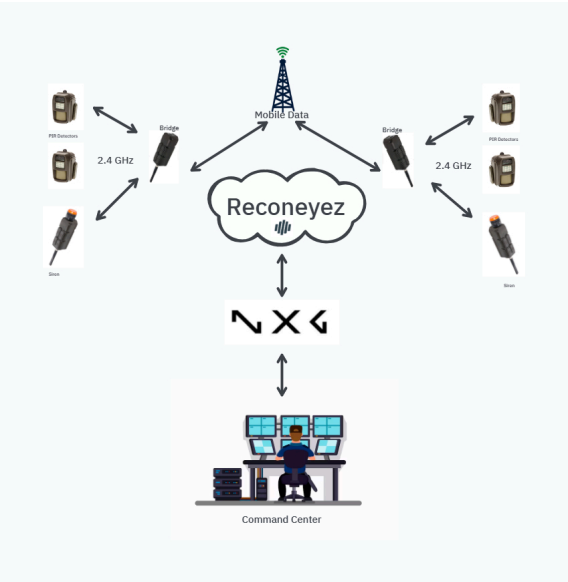
System Architecture

The architecture follows a distributed design that enables wireless operation and centralized monitoring through the Genesis platform.
The diagram below illustrates how detectors, sirens, and bridges interact within the Reconeyez and NXG ecosystem and how data flows to the command center.



Field Layer – Detection Network

At the outer layer, battery-powered **PIR Detectors** continuously monitor motion activity within their coverage area. When movement is detected, the detector captures an image for verification and transmits it over a **2.4 GHz wireless link** to the nearest **Bridge**.

Sirens are also connected within this wireless network. They receive activation commands from the bridge whenever an alarm event is confirmed, providing immediate on-site deterrence with both audible and visual alerts.

Communication Layer – Bridge Connectivity

Each **Bridge** acts as the local communication gateway for its group of detectors and sirens.
The bridge aggregates all sensor data, alarms, and device status information, then transmits it securely via **mobile data (4G)** to the **Reconeyez Cloud**.
This design allows multiple independent Reconeyez clusters to operate in the same site or region without requiring fixed network infrastructure.

Cloud Layer – Reconeyez Cloud Platform

The **Reconeyez Cloud** manages device authentication, event storage, and secure API communication.
It ensures that all captured alarms, images, and health updates are centralized and available for remote integration.
Genesis connects to this cloud platform through a secure API and FTP connection, retrieving live alarm events, device health data, and verified images.

Integration Layer – Genesis NXG

Within the Genesis ecosystem, Reconeyez appears as a unified device type.
All events, images, and device status updates from the Reconeyez Cloud are normalized into the Genesis event model.
This allows seamless alarm processing alongside other systems such as Teltonika, Victron, or EFOY within the same monitoring environment.

Monitoring Layer – Command Center

Operators at the **Command Center** receive real-time alerts through Genesis.
When a motion detection occurs, the system displays the corresponding image verification and device details.
Operators can validate the event, initiate response actions, or acknowledge alarms directly from the Genesis interface or through connected CMS platforms such as IMMIX, AmWin, or Evalink.

End-to-End Flow Summary

1. Motion detected by PIR detector.
2. Image and event sent to Bridge over 2.4 GHz.
3. Bridge uploads event to Reconeyez Cloud via 4G.
4. Genesis retrieves and classifies the event.
5. Command Center receives verified alert and initiates response.

Configuration Overview

Reconeyez integration in Genesis is achieved through a secure connection to the Reconeyez Cloud API. Once connected, the Genesis platform automatically retrieves device status, event notifications, and image data.

Configuration in Reconeyez Cloud

To integrate Reconeyez devices with the Genesis ecosystem for any new tenants, you must contact **Reconeyez Technical Support** at **support@reconeyez.com.**
Their team will activate the integration handshake required for both **Alarm Forwarding** (via SIA DC09 protocol) and **Alarm Image Forwarding** (via a secured FTP channel).

Information Required by Reconeyez Support

The following details need to be provided to Reconeyez Support to complete the integration setup:

Parameter	Description
Message data format	Must be SIA-DCS
Server IP address and port	18.158.38.178, 19231

Default account identification and device info	Account information to map Genesis receivers
Receiver Number (optional)	Rrcvr
Account Prefix (optional)	Lpref
Account Number	#acct
Group number (optional)	Defines device grouping
Zone number (optional)	Defines area or detection zone
128-bit encryption key (optional)	Currently not supported in Genesis
FTP server IP address and port	For alarm image forwarding, please check with NXG Support.
FTP user and password	Credentials for FTP upload
SSL certificate (optional)	Required only for FTPS with self-signed certificates

Area Assignment in Reconeyez Cloud

After the integration request is processed, Reconeyez will associate a specific **Area** with Genesis Cloud that supports alarm forwarding.
All new devices added in Genesis must be moved into this Area to enable alarm transmission to Genesis.

Once the device is assigned to this Area, update its **Description** field in Reconeyez Cloud with the account number (provided by Genesis) that should be used for SIA DC09 forwarding. The syntax is:

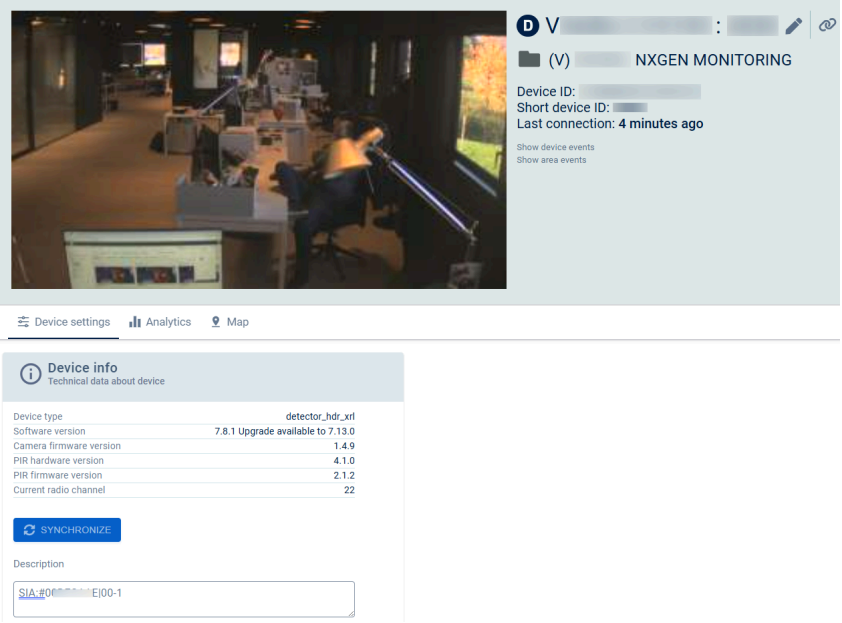
> SIA:#<Account ID>|<Area>-<Zone>

Example:

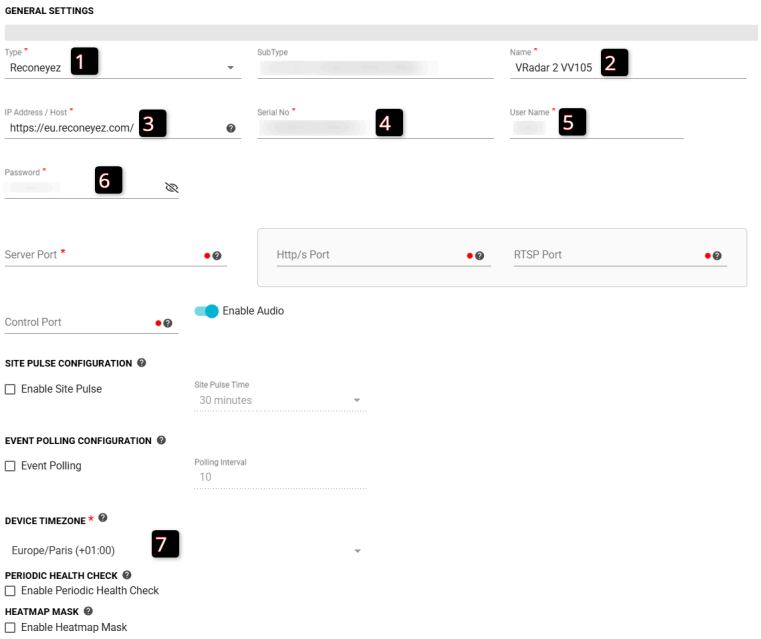
> SIA:#00DDAAEE|01-1

This indicates that the account ID `00DDAAEE` will be used when forwarding alarms to Genesis, with Area `01` and Zone `1`.

Sample Configuration screen below:



Configuration in Genesis



Parameter	Description
Type	Must be Reconeyez
Name	Unique Device name
IP Address/Host	Reconeyez cloud end point with region (ex: eu.reconeyez.com (for Europe) uk.reconeyez.com (for UK), na.reconeyez.com (for North America), au.reconeyez.com (for Australia), or demo.reconeyez.com (for demo sites)
Serial No	Unique Device ID (Refer the Reconeyez cloud portal)
UserName, Password	A valid username and password with full access permissions is required to authenticate and access the Reconeyez API suite
Device Timezone	Timezone of the device configured in Reconeyez Cloud Portal

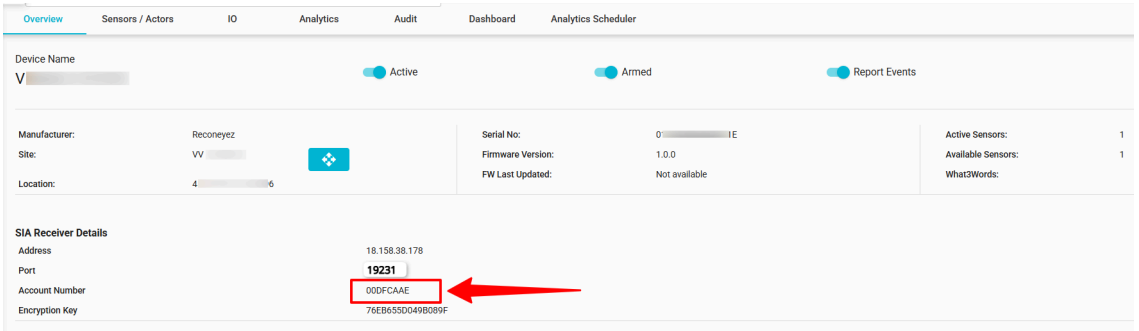
Device Discovery and Account Mapping in Genesis

After completing the configuration steps mentioned above, select **Discover** in Genesis to connect with the **Reconeyez Cloud**.
This process retrieves the list of available Reconeyez devices along with their details, including:

- Device type (e.g., detector, bridge, or siren)
- Associated area configuration from Reconeyez Cloud

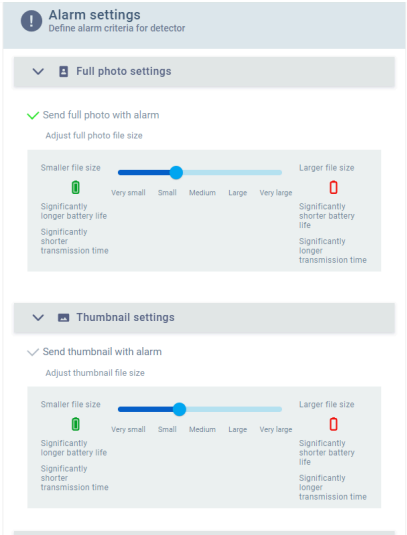
Once the devices are successfully discovered and added to Genesis, the system will automatically assign a **unique Account ID** to each device.
This account number will be used in all subsequent alarm messages received from Reconeyez for accurate mapping within Genesis.

(Refer to the screenshot below for an example of the Discovery and Account Assignment screen.)



Alarm Configuration

Goto Alarm settings in the Detector Device Settings

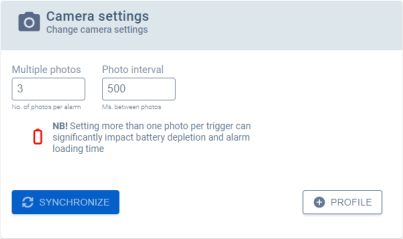


Ensure that the **photo size** configuration in the Reconeyez Cloud is set to a **reasonable level** — preferably between *Small* and *Medium*.
This balance ensures that the **detector's battery life** is preserved while still maintaining **sufficient image quality** for Genesis to accurately analyze alarm frames and apply **false alarm filtering**.

additionally, to ensure Genesis receives sufficient visual frames for reliable alarm verification, adjust the camera parameters as follows:

- **Multiple Photos:** 3
- **Photo Interval:** 500 milliseconds

This configuration helps Genesis analyze multiple alarm frames, improving accuracy in differentiating between false and real alarms.



Integration Features Overview

Feature	Description
Alarm Routing and Processing	Alarms are received via **SIA DC-09** and automatically mapped to the corresponding account and site in Genesis. Event type, priority, and timestamp are parsed for operator routing and automation rules.
Photo / Video for Verification	Alarm images and short video clips (if possible) are retrieved via a **secured FTP/FTPS** channel and displayed in Genesis for operator verification and audit trails.
Media URL for Verification	Generates and stores a **Reconeyez media URL** for every verified alarm to allow quick access, reference, and external sharing through Gene
Remote Arming / Disarming	Allows site or device-level arming/disarming via Genesis, synchronized with Reconeyez Cloud schedules and permissions.
Device Status Monitoring	Retrieves device connectivity, signal strength, and battery health from the Reconeyez Cloud for live status visibility and fault notifications i Genesis.
Map and Geofence Integration	Device coordinates are visualized on Genesis maps with support for defining geofences to trigger events or analytics.
Remote Siren Activation	Enables operators to activate sirens connected to Reconeyez bridges directly from the Genesis Map Interface or automated rules.
Remote Snapshot Request	Allows operators to trigger an immediate snapshot from any active detector for manual verification. Media retrieved through the secure FTP channel.

Summary

The Genesis integration with Reconeyez provides a complete, low-power, and cloud-connected perimeter protection solution.
It combines motion-based visual verification with intelligent alarm management, ensuring reliable protection for remote or temporary sites.

With proper configuration and alarm rule tuning, Reconeyez becomes a seamless part of the Genesis ecosystem, delivering verified intrusion alerts, continuous device health monitoring, and improved situational awareness across all connected locations.